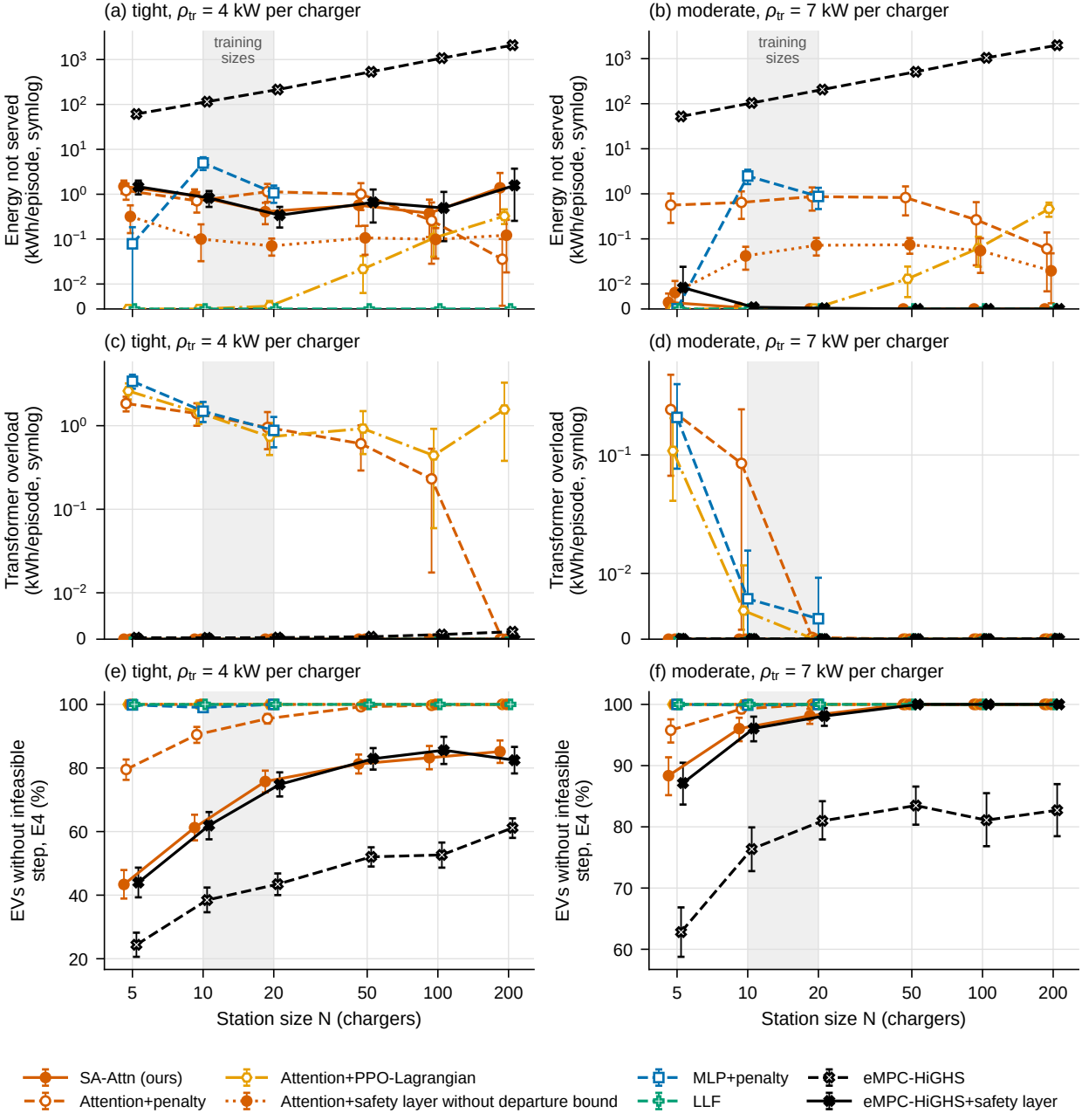
(a) tight, ρ_{tr} = 4 kW per charger
(b) moderate, ρ_{tr} = 7 kW per charger
(c) tight, ρ_{tr} = 4 kW per charger
(d) moderate, ρ_{tr} = 7 kW per charger
(e) tight, ρ_{tr} = 4 kW per charger
(f) moderate, ρ_{tr} = 7 kW per charger
training sizes
Energy not served (kWh/episode, symlog)
Transformer overload (kWh/episode, symlog)
EVs without infeasible step, E4 (%)
Station size N (chargers)
SA-Attn (ours)
Attention+penalty
Attention+PPO-Lagrangian
Attention+safety layer without departure bound
MLP+penalty
LLF
eMPC-HiGHS
eMPC-HiGHS+safety layer